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Advanced Java Interview Scenarios for Job Switchers with Common Mistakes (3+ Yrs Experience)

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Scenario 1: Deadlock in Multithreading

- **Topic:** Concurrency & Synchronization
- **Scenario:** Two threads are trying to acquire two locks (A & B). Thread-1 locks A then waits for B, while Thread-2 locks B and waits for A. Deadlock occurs.
- **Solution:**
  - Always acquire locks in a fixed global order.
  - Use tryLock() with timeout from ReentrantLock.
- **Common Mistake:** Using synchronized blocks without lock ordering.
- **Follow-up Question:** How would you detect deadlocks in production?

Scenario 2: Optimizing a REST API Under Heavy Load

- **Topic:** Performance + Spring Boot
- **Scenario:** Your API processes 10K requests/second, but response time spikes > 3s.
- **Solution:**
  - Use connection pooling (HikariCP).
  - Cache frequently accessed data with Redis.
  - Use async controllers (@Async, CompletableFuture).
- **Common Mistake:** Blocking I/O calls, no caching.
- **Follow-up Question:** How would you implement backpressure?

Scenario 3: Handling Huge Data with Streams

- **Topic:** Java Streams + Memory Management
- **Scenario:** You need to process a 10GB CSV file with filtering + aggregation.
- **Solution:**
  - Use Stream.of(File.lines()) with parallel streams.
  - Consider batch processing + Spliterator for custom iteration.
- **Common Mistake:** Collecting the whole dataset into memory (.collect() on huge lists).
- **Follow-up Question:** When would you choose parallelStream() vs reactive frameworks like Project Reactor?

Scenario 4: Retry Logic for External API Calls

- **Topic:** Resilience + Java + Microservices
- **Scenario:** Your service depends on an unreliable 3rd-party API (timeouts, 502 errors).
- **Solution:**
- Implement exponential backoff retry using Spring Retry or Resilience4j.
- Use Circuit Breaker to prevent cascading failures.
- **Common Mistake:** Infinite retries → thread exhaustion.
- **Follow-up Question:** Difference between Retry vs Circuit Breaker?

Scenario 5: Preventing SQL Injection in Java Apps

- **Topic:** Security + JDBC/JPA
- **Scenario:** A junior dev concatenates user input into SQL query.
- **Solution:** Always use Prepared Statements (? placeholders).
- **Common Mistake:** String concatenation in queries, ignoring ORM protections.
- **Follow-up Question:** How can SQL injection still happen even with Hibernate?

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